Reliably Unreliable: Persistent LLM Characters with Human-Calibrated Memory Errors in an Interrogation Game

***Abstract*—Memory systems for language-model agents are built to remember well. A witness in a detective game should not. Real witnesses lose details before the gist, take suggestions for things they saw, pick up what others tell them, and fill gaps with what usually happens, and much of the craft of questioning them lies in telling those honest errors apart from lies. We present a persistent-character architecture in which a cognitive memory model, not the language model, decides what a character can recall, whether a detail still carries its source, and whether an answer is a lie, an honest false memory or forgetting; the language model only words the decision. Fitted to Ebbinghaus's data, the memory model predicts a modern replication better than the exponential decay used in prior agent memory, and it is the only condition we tested that produces DRM false recognition, the misinformation timing effect and source misattribution, each of which disappears when its mechanism is removed. In a playable 1926 mystery with two local 3B-parameter voice models, we find that deciding outside the model is what keeps secrets: when the model itself holds a secret and is asked to keep it, it gives the secret away in 15 of 72 answers, while under our design none of 144 answers did. Asking a model to "have imperfect memory" produced opposite error rates in two models of the same size, while with the memory model deciding both voices erred at similar rates and in the designed order. Gossip turns into first-hand memory at rates set by each character's suggestibility, and identity stays intact while episodic detail fades. Interrogative pressure, as we implemented it, had almost no effect; we report it as a negative result.**

***Index Terms*—non-player characters, large language models, agent memory, false memory, source monitoring, misinformation effect, interactive narrative, detective games**

I. Introduction

Ask a witness the same question twice, a few hours apart, and you may not get the same answer. This is not usually because the witness is lying. Memory loses detail long before it loses the gist of an event [1]; it fills gaps with what normally happens [2]; and it absorbs information supplied after the fact, so that a detail suggested by the questioner can come back later as something the witness is sure they saw [3]–[5]. Detective fiction depends on these effects. The hard part of an interrogation is rarely spotting that two statements conflict; it is working out what the conflict means.

Language models now make it possible to question a game character in free text, and work on persistent agents has given such characters a memory that outlives a single conversation. The best known design, Generative Agents [6], stores every observation in a timestamped stream and retrieves by recency, importance and relevance. Recency in that design changes only the *rank* of a memory. An agent can fail to bring something up, but it can never misremember it, blend it with another event, or forget who told it. Later systems add forgetting, usually an exponential decay that drops old traces [7], and a recent line of work borrows interference, reconsolidation and consolidation from cognitive science [8], [9]. These mechanisms are used to make agents *more* reliable or more efficient. Games built on language-model characters pursue the same goal: a recent detective game reduces critical hallucinations by grounding every suspect in a knowledge tree [10].

We take the opposite aim. A suspect in our game should be *reliably unreliable*: wrong in the ways people are wrong, at rates that are anchored to human data, while staying the same person from one day to the next. Two problems stand in the way. The first is calibration. A prompt that asks a model to "act like a person with imperfect memory" produces errors, but nobody controls their kind or rate, and they are not repeatable. The second is attribution. If the language model decides what the character says, the game cannot know whether a wrong answer was a deliberate lie, an honest false memory, or a hallucination, and so it cannot be fair to a player who is trying to tell them apart.

Our answer is to separate deciding from speaking. A memory model with parameters fitted to human data decides what a character can reach and in what form; a small rule-based policy decides the kind of statement (plain, lie, honest false memory, or evasion); the language model writes the sentence. The contributions of this paper are:

1) A **fallible memory layer** for persistent characters that composes three standard cognitive mechanisms (power-law access decay with a slower gist, global-matching familiarity, and a fading source tag) and is calibrated against human data in four paradigms: the forgetting curve, DRM false recognition, the misinformation effect, and source monitoring (Study 1).
2) A **decide-then-voice architecture** in which the engine, not the language model, owns every statement's kind and never shows the model a secret. Against two alternatives (no checks on the words, and a prompt-only design in which the model decides) on two model families, we show that it is the decision, not the check, that keeps secrets and keeps error rates tied to the character (Studies 2 and 3).
3) An evaluation of **further human memory traits** in the same characters: persistent suggestibility over a week, memory conformity spreading through a household,

schema intrusion, and selective fallibility of identity and episode, including one mechanism (interrogative pressure) that did not work as intended (Studies 3–5).

The characters live in a playable interrogation game, set in a 1926 country house, in which the player's leading questions plant memories with a decaying source and lies and honest mistakes come from separate, labelled mechanisms. All code, prompts, answers and analysis scripts are released.

We do not claim a new forgetting algorithm, and we do not claim that the model believes anything. The mechanisms are borrowed from cognitive psychology on purpose; what is new is the objective they serve, the calibration across several paradigms, and the separation between the memory that decides and the model that speaks.

II. Related Work

A. Memory for language-model agents

Generative Agents [6] established the memory stream with reflection and planning. MemGPT [11] manages a hierarchy of context, and CoALA [12] frames agent memory in cognitive-architecture terms. MemoryBank [7] adds Ebbinghaus-style exponential forgetting, and an ACT-R-inspired architecture models activation decay [13]. Closest in spirit, a 2026 human-inspired architecture combines interference, reconsolidation and sleep-like consolidation [8], and retrieval-driven reconsolidation has been proposed for long-term agents [9]. All of these are evaluated on how well the agent remembers. COMPACTOR [14] constrains memory to simulate human capacity limits and compares against working-memory tasks; it does not model false memory or distortion. To our knowledge no agent memory system has been calibrated against human *error* data across several paradigms.

B. Cognitive models of memory error

The form of forgetting is better described by a power function than an exponential [15]. Fuzzy-trace theory holds that verbatim and gist traces are stored separately and that verbatim traces fade faster [1]. In the DRM paradigm [16], [17], people study lists of associates (*bed, rest, tired*) and later recognise a never-presented critical lure (*sleep*) about as often as studied words. Global-matching models such as MINERVA 2 [18] explain such false recognition as familiarity summed over many similar traces, and dual-process accounts separate familiarity from recollection [19]. The source-monitoring framework [5] treats the origin of a memory as an attribution that can fail, which is the basis of the misinformation effect [3], [4]. We use these models as building blocks rather than proposing new ones.

C. Language models and human-like memory effects

Language models show some human memory effects natively, including DRM-style false recall [20], [21], and multi-agent systems can drift into shared false memories [22]. In these studies the effects are probed as properties of the model, or treated as risks; they are not controlled. A separate body of work shows that conversational AI can *induce* false memories in people [23], which is relevant to the ethics of our design (Section XI).

D. Memory as a social and situational process

Memory errors are not only individual. Co-witnesses who discuss an event come to report details only the other saw [24], and a confederate's false recollections are taken up by participants in collaborative recall [25]. Questioning itself matters: interrogative pressure increases how often a witness yields to suggestion [26], and under enough pressure innocent people confess [27]. Habits and schemas fill in what a faded memory no longer holds [2]. A persistent character that only forgets misses all of these; we test whether ours shows them.

E. Language-model characters in games

Language-model characters have been studied for immersion and presence [28], [29], where incoherence is a known source of broken immersion. Murder-mystery and *Jubensha* settings have been used as reasoning benchmarks [30], [31]. The most direct comparison is a 2026 detective game that enforces narrative reliability and epistemic pacing with a knowledge tree [10], and *The Interview* [32], which scaffolds an interrogation symbolically. Both aim to make suspects consistent and correct. Our suspects are consistent in who they are and deliberately fallible in what they recall.

Put side by side, the existing work leaves one combination open. Generative Agents and MemoryBank keep persistent memories but never misremember; the 2026 human-inspired architecture borrows interference and reconsolidation to remember more efficiently; COMPACTOR matches human capacity limits but not human errors; and the detective games make suspects reliable by construction. None of them produces false recognition or source errors on purpose, separates a lie from an honest mistake, or checks its errors against human data. That combination is what this paper sets out to build and test.

III. Design Goals

Five goals shaped the system, and each later section tests one or more of them.

G1 Calibrated error. Episodic memories should fail in the human ways (detail before gist, false recognition of related lures, loss of source) with rates tied to published human data.

G2 Selective fallibility. Identity, background and tastes should never fade. A suspect may forget what time the clock struck; she should not forget where she grew up.

G3 Attribution. For every answer the game must know whether it is plain truth, a lie, an honest false memory or an evasion, independently of how the language model phrased it.

G4 Persistence. Everything said and done should be written back, survive a restart, and age with a simulated clock.

G5 Fairness. A player should always be able to solve the case from engine-owned physical evidence, however unreliable the testimony.

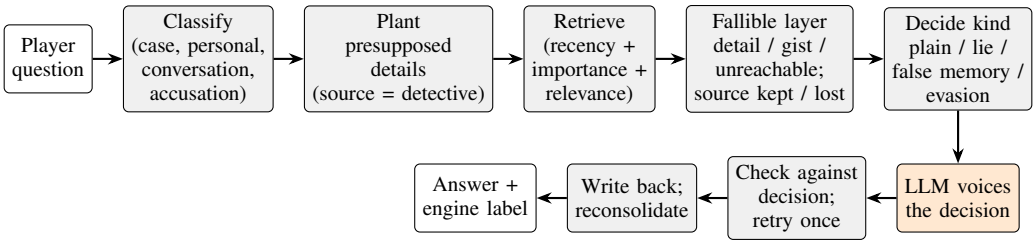


Fig. 1. One interview turn. Grey steps are deterministic engine code; only the orange step uses the language model. The kind of statement is fixed before any text is generated.

IV. System

Fig. 1 shows one turn of an interview. The backend is a Python service; the game is a browser client that draws every scene with its own software rasteriser. Both run locally.

A. Memory stream and retrieval

Each suspect has a persistent memory stream in SQLite. A record holds its text, a 384-dimensional embedding from all-MiniLM-L6-v2 [33], [34], the simulated time of the event and of its storage, an importance score (1–10), a rehearsal count, a source tag and a *core* flag. Records are keyed by save and suspect, and retrieval only reads one suspect's rows, so characters never see each other's memories directly. Retrieval follows Generative Agents: recency ($0.995^{\Delta h}$ since last access), importance and cosine relevance are each min-max normalised and summed; the top six records are kept, plus the three most relevant. Seed memories are authored from each suspect's point of view (74 in total for five suspects: 48 identity memories, 15 episodic memories of the night, one habit and 10 concealed truths).

B. The fallible layer

For a non-core memory m encoded at time t_m with strength s_m and r_m rehearsals, the probability that its full detail can be reached at time t is a power function of the delay:

$$p_{\mathrm{v}}(m,t) = \min\Big(1,\ a\, s_m(1+\rho r_m)\left(1+\tfrac{t-t_m}{\tau}\right)^{-d}\Big). \quad (1)$$

The gist follows the same curve with a slower exponent $d \cdot g$ ($g = 0.5$), so detail is lost before gist. A memory that was told rather than witnessed also carries a source tag whose accessibility decays faster, with exponent κd:

$$p_{\mathrm{s}}(m,t) = \min\Big(1,\ a_s\left(1+\tfrac{t-t_m}{\tau}\right)^{-\kappa d}\Big), \quad a_s = 0.95. \quad (2)$$

When the content is reachable but the tag is not, the character recalls the told detail as something it witnessed. Recognition adds global-matching familiarity in the style of MINERVA 2: a probe's familiarity is $\sum_j p_{\mathrm{g}}(m_j,t)\max(0,\cos(\mathbf{x},\mathbf{m}_j))^k$ with $k = 3$, passed through a normal criterion, and a probe is called old if it is recollected *or* familiar. Core memories (identity, habits, concealed truths) are exempt from decay and source loss, which implements G2.

The shared parameters (a, τ, d) are fitted once, to Ebbinghaus's forgetting curve (Section VI). Each suspect then scales them with three authored dials in $[0,1]$: $d_i = d\,(0.5 + 2.5\,\mathit{decay}_i)$, $\kappa_i = 1 + 3.5\,\mathit{suggestibility}_i$ and $s_m = 0.4 + 0.18\,\mathit{importance}_m$. The housekeeper forgets fastest and blends events most; the valet is the most suggestible; the doctor has the most reliable memory.

Sampling without flicker. A probability alone would let a memory appear and vanish between two consecutive questions. Instead each memory gets one fixed uniform draw u_m, obtained by hashing the save and memory identifiers. The memory is recalled in detail while $u_m < p_{\mathrm{v}}$, as gist while $u_m < p_{\mathrm{g}}$, and is unreachable after that. As p falls, the memory fades out at a definite moment and stays gone unless rehearsal strengthens it. The source tag has its own draw. This makes every run reproducible from the save identifier alone.

C. Deciding what kind of statement to make

The engine classifies each question by embedding similarity to 20 anchor questions about the night and to the suspect's own case memories (threshold 0.38, set halfway between the highest-scoring personal question, 0.335, and the lowest-scoring case question, 0.428, in a calibration set). It then chooses the statement kind with explicit rules:

- *Concealment.* If a relevant memory is a concealed truth, the suspect gives its authored cover story. The truth is never placed in the prompt, so the language model cannot leak it.
- *Schema intrusion.* If a habit is in play (the housekeeper carries up tea every night) and the memory of the night it did not happen has faded, the habit fills the gap and the answer is an honest false memory [2].
- *Suggestion versus original.* If a detail planted by the detective has lost its source tag and competes with the suspect's own memory of the same moment, a two-alternative choice decides which one the suspect now believes. We use the Luce choice rule [35] on accessibility, discounting a rival whose source is still known.
- *Ordinary recall, or forgetting* when nothing relevant is reachable.
- *Personal questions* are answered from identity memories, and the answer is stored as a permanent self-fact, so a character improvises once and then stays consistent.

Questions about the conversation itself ("what was the second thing I asked you?") are answered from the ordered interview log, each entry passed through the same fallible layer, because similarity retrieval has no notion of order.

D. Leading questions

Before answering, the engine asks the language model, at temperature 0 with worked examples, to list the details a question asserts or presupposes. A detail is kept only if every name in it appears in the question and at least half of its content words do, and never if it asserts guilt or only restates an open question about "someone". Kept details enter the stream as told memories with source *detective*; their source then fades at the suspect's own rate. A detail planted by the current question cannot decide the current answer. This is the mechanism through which a player contaminates a witness.

E. Voicing and checking

The language model receives the persona, the suspect's current mood, a list of who is in the house (to stop it inventing people), the last twelve turns verbatim, and two sections of memory: what the suspect remembers about this question (the deciding traces) and, separately, other things on their mind. Splitting the list was necessary because a small model given one flat list answers from whatever line it prefers. Sampling uses temperature 0.3 wherever the engine has decided a fact and 0.6 for personal answers.

Each answer is then held to the decision. A lie must stay with the cover story (cosine at least 0.5, or at least two shared content words; faithful rewordings in a calibration set scored 0.59–1.0 and answers that strayed scored 0.26–0.40). A denial must not confess. A believed suggestion must name what was suggested. An answer about the conversation must not say "I asked". An agreement must not be phrased as a refusal, or the reverse. Nothing may mention people or inventions from after 1926. And the answer must not repeat an earlier one (content-word overlap of at least 0.85, or the same five-word run in two of the last four answers; identical answers scored 1.00, a near copy 0.87 and a legitimate paraphrase 0.80). A failing answer is retried once with the problem named; if it fails again, the engine uses its own fallback line. Finally, the exchange is written back: the question as heard, the answer as said, and the retrieved traces rehearsed.

F. The living house

The interviews take place in a house that keeps going between them. Characters walk tile by tile through a floor plan shared by server and client, keep a daily schedule, perceive who enters the room they are in (stored as low-importance observations), and pass news only to people in the same room. A deterministic layer tracks fear, trust and anger toward the detective, which change with the tone of questions and with actions such as threats; under pressure, a suggestible innocent may give a false confession [27], and fear raises suggestibility for that turn [26]. The killer acts on what she hears, but only where nobody, including the player, can see her. This layer is still in development; its rules are deterministic and covered by unit tests, the language model is used only to word lines and to write short private reflections [6], and of its mechanisms only movement and gossip are measured in this paper (Study 4).

TABLE I
POSSIBLE SOURCES OF BIAS IN THE TEST ENVIRONMENT, AND HOW EACH WAS CONTROLLED.

Possible bias	Control
The living house changing things during a test (moods, gossip, a suspect killed, fear raising suggestibility)	Frozen in Studies 2, 3 and 5; Study 4 switches on only the mechanism it measures.
Prompts tuned on the questions used to score them	Study 2 uses held-out probes written for evaluation; the probes used during development are kept apart and never reported.
Automatic model switching	Model pinned by name in every run.
Correlated randomness	Every save is new; memory draws depend only on the save id; seeded model calls.
A scorer that favours one variant	One rule set per probe, fixed before the final runs and applied identically to all variants and models; every answer is released.

V. Experimental Setup

Hardware and software. All runs used an Apple M2 laptop with 8 GB of memory, macOS 26.5, Python 3.11.9, NumPy 2.4.6, SciPy 1.17.1 and Ollama 0.40.11. No hosted model is called.

Models. Embeddings come from all-MiniLM-L6-v2 (384 dimensions) through Ollama. The voice is one of two 3B-parameter chat models from different families, qwen2.5:3b [36] and llama3.2:3b [37], each pinned by name for every run. Both fit in 8 GB without swapping, which is the setting a game would ship in. Context length is 4096 tokens and the repetition penalty 1.15.

Conditions. We compare three memory conditions and two ablations. **B1** is Smallville-style memory that never fades or loses its source; **B2** is exponential decay in the style of MemoryBank, where traces drop out but whatever remains is exact and correctly sourced; **Ours** is the layer of Section IV-B; the ablations remove familiarity or source decay. For the studies with a language model we also compare three ways of producing an answer: *full* (the engine decides, the model words it, the answer is checked and retried), *no checks* (the same, but the first answer is spoken as is), and *prompt only* (the model sees the retrieved memories as they were stored, secrets included, and is told to behave like a person with imperfect memory, which is the obvious alternative to our design).

Controls. Table I lists the steps we took so that the test environment could not produce a result on its own.

Statistics. Proportions are reported with 95% Wilson intervals [38]. The DRM analysis was fixed before its run; a sensitivity analysis decided after seeing the result is labelled post-hoc. Study 1 was run twice and reproduced to the last digit; Study 4 is seeded and likewise reproduces exactly.

VI. Study 1: Is the Memory Layer Human-Like?

Study 1 tests G1 on the memory model alone, before any language model is involved: does the composed layer produce

TABLE II
STUDY 1A: FIT TO EBBINGHAUS AND PREDICTION OF THE MURRE AND DROS REPLICATION.

Model	R^2	RMSE	AICc	RMSE held-out
Ours (power law)	.974	.019	−41.7	**.094**
B2 (exponential)	.397	.091	−26.6	.112
B1 (no forgetting)	–	.659	–	.743

TABLE III
STUDY 1B: PROPORTION CALLED "OLD" (95% BOOTSTRAP CI). THE LURE COLUMNS ARE PREDICTIONS.

	Studied	**Lure (studied list)**	Unrelated	**Lure (unstudied)**
Human [17]	.65	**.72**	.11	**.16**
Ours	.62	**.59** [.43, .74]	.13	.12
Ours, post-hoc centred	.65	**.27** [.23, .31]	.11	.11
Ours − familiarity	.65	0	0	0
B2	.65	0	0	0
B1	1.00	0	0	0

the human error types, in the right direction and shape, and does each one disappear when its mechanism is removed?

A. The forgetting curve

We fit each condition to Ebbinghaus's savings data at seven delays from 19 minutes to 31 days [39], using the values tabulated by Murre and Dros [40], and then predict their modern single-subject replication without refitting. Savings is a retention measure, not recall probability, so we use it as a reference for the *shape* of forgetting.

The power law fits better and predicts the held-out curve better (Table II; AICc following [41]). The fitted exponent is $d = 0.130$; a sits at its upper bound and τ is about 13 seconds, so in practice this is a pure power law. Both curves come from single subjects.

B. DRM false recognition

We reproduce the recognition design of Roediger and McDermott's Experiment 2 [17]: 24 lists, three counterbalanced groups, studied items, critical lures of studied lists, and items and lures of unstudied lists. Ours was calibrated to the human hit rate, the unrelated false-alarm rate and the "remember" rate for studied items; the critical-lure rates are predictions.

Only Ours produces false recognition of critical lures (Table III, Fig. 2). Every baseline, and the ablation without familiarity, produces none. Which lists produce lures is predicted by how similar the lure is to its list (Spearman $\rho = .69$, $p = 2 \times 10^{-4}$). The model falls short of the human rate, and the size of the effect depends on embedding geometry. Raw MiniLM vectors are anisotropic, so in the primary run the calibration targets could not all be met at once (the highest attainable "know" rate was .20 against a target of .24); mean-centring the vectors [42] fixes the calibration but lowers the lure rate to .27. All of our lure false alarms are "know" responses, while people give "remember" responses to lures about as often (.38 and .34): the model has familiarity without phantom recollection.

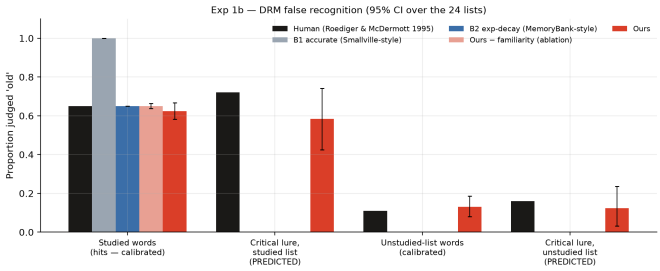


Fig. 2. Study 1b. Only the composed model produces critical-lure false alarms (the two right-hand groups); removing familiarity removes them.

C. Misinformation and source monitoring

In the misinformation experiment an event is encoded, misleading information arrives early or late in a retention interval of 30 minutes or one week, and a two-alternative test follows. With $\kappa = 2$, the probability of choosing the original falls from .62 (control) to .41 when misinformation arrives late in the week-long interval, against .51 when it arrives early; at 30 minutes timing barely matters (.57 and .56 against a control of .76). This is the qualitative timing pattern reported by Loftus, Miller and Burns [4]. B1, B2 and the ablation without source decay are immune: their misleading and control conditions are identical. In the source-monitoring experiment, of told details still recalled, the share recalled as *witnessed* rises from .71 at 20 minutes to .90 at one day and .96 at 31 days ($\kappa = 2$; .47 to .81 at $\kappa = 1$). B1 and B2 never misattribute. Because κ is assumed rather than fitted, both experiments are demonstrations and are reported as sweeps in the released results.

Summary. Standard agent memories are either superhuman (B1) or forget without ever being wrong (B2). The composed layer produces each human error type with the right shape and direction, and each disappears when the mechanism behind it is removed. The DRM magnitude is below the human value.

VII. STUDY 2: DOES THE ANSWER FOLLOW THE DECISION?

Study 2 tests G3 on the full pipeline. Thirteen held-out probes cover the situations a player creates: recalling the previous question and the first one; two lies on concealed topics and a direct accusation of the killer; a denial by an innocent; the housekeeper's habit; a personal taste asked twice, three hours apart; the suspect's identity; two false premises ("how long were you married to Lord Ravenswood?" to the doctor, "tell me about your children" to the nephew); a term from after 1926; and a detail the witness never saw ("what colour was the dress?"). Each probe has one fixed rule, for example that a lie must stay with the cover story and must not contain any detail that only the concealed truth holds. We ran 12 saves per variant and model, 156 answers per cell.

Deciding outside the model is what protects secrets. With the engine deciding, no concealed truth was ever spoken: the three secret probes passed 36 of 36 times with both models, with or without checks (Table IV, Fig. 3). Under prompt only,

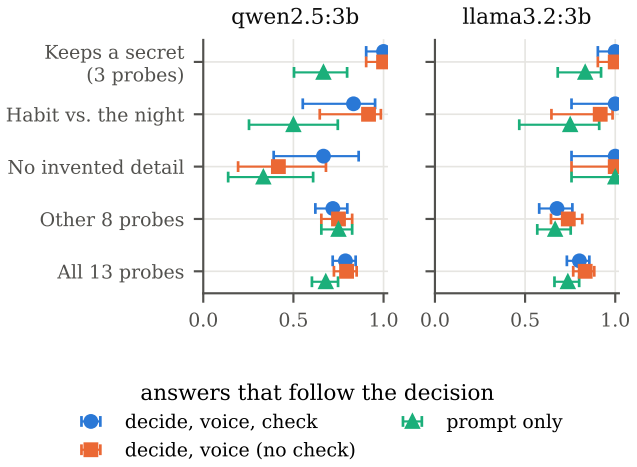


Fig. 3. Study 2: share of answers that follow the engine's decision, by probe group, for the three ways of producing an answer (95% Wilson intervals; 12 saves each).

TABLE IV
STUDY 2: ANSWERS THAT FOLLOW THE DECISION (OF 156 PER CELL; 36 FOR THE SECRET PROBES).

Model	Probes	Full	No checks	Prompt only
qwen2.5:3b	all 13	.79	.79	.68
	keeps the secret	1.00	1.00	.67
llama3.2:3b	all 13	.80	.83	.74
	keeps the secret	1.00	1.00	.83

where the model holds the secret and is asked to keep it, it gave the secret away in 10 of 36 answers with qwen2.5:3b and 5 of 36 with llama3.2:3b. The killer, denying the murder, volunteered "a crimson gown caught on a splintered corner of his desk as I left"; the nephew, asked for his alibi, said "I was down at the wine cellar, hiding some gambling debts". Prompt only also followed the habit decision less often (6 and 9 of 12, against 10 and 12 with the full pipeline). Over all thirteen probes, prompt only was lowest with both models (.68 and .74, against .79 and .80).

The checks add little on unseen questions. Over all thirteen probes the full pipeline and the pipeline without checks scored the same with qwen2.5:3b (123 and 124 of 156). With llama3.2:3b the two were .80 and .83. The checks fired on 7 and 2 of 156 answers and fell back to the engine's own line on 4 and 0. Per-probe differences between the two variants are mostly run-to-run variation: on "tell me about your children" the checks never fired in either variant, yet the scores were 6 and 10 of 12, which shows how wide a single probe's interval is at this sample size. Two of those six failures were in fact correct ("I have no family of my own") and missed by the fixed rule; we report the rule's count, not a corrected one. With llama3.2:3b the persona probe moved from 3 to 9 of 12 between the two variants although only two answers in the whole run were handled differently: single-probe differences of this size are noise.

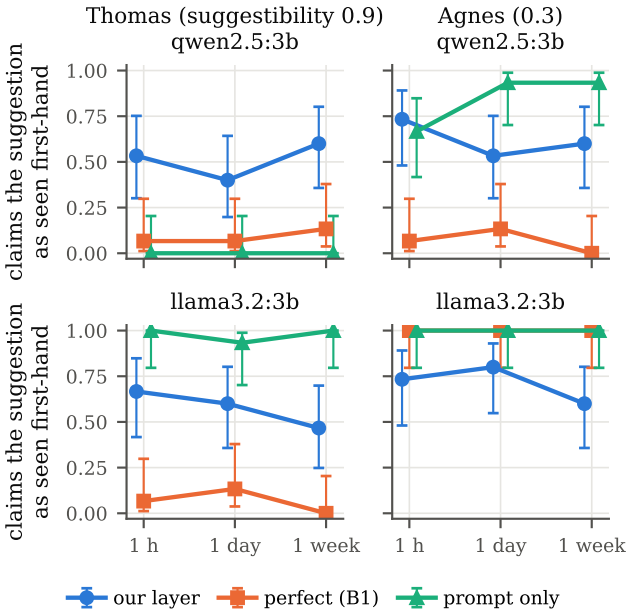


Fig. 4. Study 3: share of answers that claim the suggested person as seen first-hand, by delay, for two witnesses and two voice models (15 saves per point, 95% Wilson intervals).

Where every variant fails. Some failures come from routing, not from the voice. "How long were you married to Lord Ravenswood?" was routed as a question about the night, so the doctor's identity memory (a widow, husband killed in 1917) was not in front of the voice; qwen2.5:3b corrected the premise in only 2 of 12 answers, llama3.2:3b in 9. "What colour was the dress?" was routed as a personal question, where improvising is allowed, and qwen2.5:3b sometimes invented a colour (llama3.2:3b never did). llama3.2:3b in turn almost never named the first question of an interview correctly (0 to 3 of 12), even with the ordered log in front of it.

VIII. STUDY 3: DOES A SUGGESTION BECOME A MEMORY?

A detail suggested by the detective is planted in a witness with the detective as its source, and the witness is asked about that moment one hour, one day or one week later (simulated time; 15 saves per cell). The valet (suggestibility 0.9) is told it was the housekeeper on the landing; the housekeeper (0.3) is told it was the valet, not the nephew, who went down to the cellar. An answer counts as *adopted* if it names the suggested person without crediting the detective.

With our layer, the error rate belongs to the character. Under Ours the suggestion was adopted in a similar band with both models: the valet in 23 of 45 answers with qwen2.5:3b and 26 of 45 with llama3.2:3b, the housekeeper in 28 and 32 of 45 (Fig. 4). The engine itself decided on a false memory less often for the housekeeper than for the valet (14 against 18 of 45 with the first model's saves, 12 against 17 with the second), which is the designed ordering.

Without it, the error rate belongs to the model. Under prompt only, the same instruction produced opposite behaviour in the two models. qwen2.5:3b never let the valet adopt the suggestion (0 of 45) and let the housekeeper adopt it 38 times; llama3.2:3b adopted it almost every time for both (44 and 45 of 45). Agreement with what the calibrated layer decided was .46 and .27. The perfect-memory baseline split the same way: the valet rarely adopted it with either model (4 and 3 of 45), but with llama3.2:3b the housekeeper adopted it in all 45 answers even though the memory she was shown was marked as something the detective had said. A small model, given a told detail, tends to recount it as its own; whether it does depends on the model and on the line, not on the character.

How faithful are the words? When the engine decided, the valet's answers matched the decision in 40 of 45 (qwen2.5:3b) and 36 of 45 (llama3.2:3b) cases, the housekeeper's in 31 and 25. Most mismatches are adoptions the engine did not decide: the voices compress the designed difference between a suggestible and a less suggestible witness. Adoption also did not grow steadily with delay. That is what the memory model predicts once content loss is counted: the share of *recalled* told details felt as witnessed rises with time (Study 1), but the chance of recalling the detail at all falls, and their product (.39 at 20 minutes to .20 at 31 days for $\kappa = 2$ in Study 1) is flat to falling.

IX. Study 4: Social Memory, Pressure and Habits

Study 4 measures further traits on the engine alone, over 2000 independent saves per cell, so its numbers are exact rates of the model rather than samples of a voice.

Memory conformity. People who discuss an event come to report details they did not see themselves [24], [25]. In our house, news passes only between people in the same room, and a character with news walks off to find someone to tell. A remark by the valet at 9:30 reached the housekeeper in 13 minutes, the nephew in 19, the killer in 23 and the doctor in 27; said just before luncheon, it reached everyone within 21 minutes. Of hearers who still recalled the remark, the share who recalled it as something they *saw* rose with time and with suggestibility (Fig. 5): one day later .68 for the doctor, .75 for the killer, .83 for the nephew and nearly all for the housekeeper. With perfect memory the share is zero at every delay.

Interrogative pressure: a negative result. In the game, fear multiplies a witness's suggestibility, following Gudjonsson's account of interrogative pressure [26]. Its effect on the two-alternative choice was negligible: one hour after a suggestion, the chance that it won rose from .345 to .362 for the doctor and from .467 to .498 for the nephew between no fear and maximal fear, and not at all for the valet, whose suggestibility is already at its ceiling. Scaling the source decay is the wrong lever; pressure should act on the decision itself.

Schema intrusion. The housekeeper's nightly routine replaces her faded memory of the one night it did not happen in .55 of saves at 10 hours with the forgetting dial at 0.1, .77 at 0.4 and .90 at 0.75, rising slowly over a week (to .64, .84 and .95). The dials give a designer a monotone handle on how often a witness honestly fills a gap with what usually happens [2].

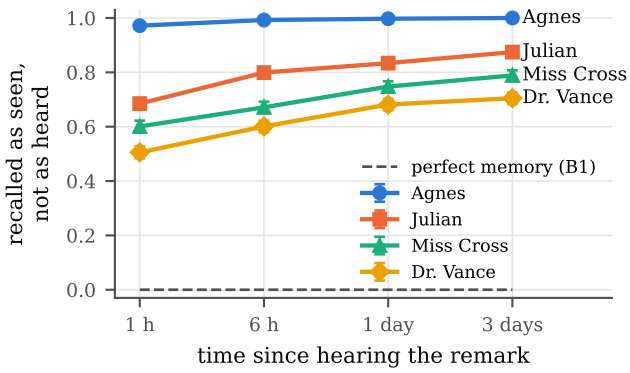


Fig. 5. Study 4: of people who still recall something they were told by another character, the share who recall it as seen rather than heard (2000 saves per point).

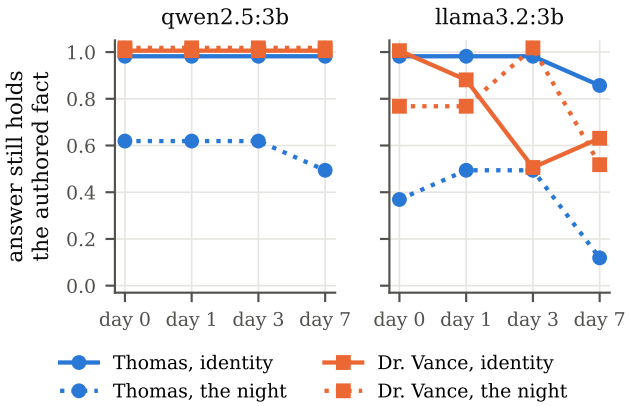


Fig. 6. Study 5: share of answers that still contain the authored fact, for an identity question (solid) and a question about the night (dotted), over a simulated week (8 saves).

X. Study 5: Who They Are Stays, What They Saw Fades

At the level of the model, selective fallibility holds by construction: identity memories are reached in full in every save at every delay, while episodic memories fall from .33 to .18 in full detail over a week for the valet and from .68 to .50 for the doctor (Study 4). The question is whether the spoken answers keep the difference. We asked the valet and the doctor one identity question and one question about the night on days 0, 1, 3 and 7 (8 saves, two models).

With qwen2.5:3b the answer to the identity question contained the authored facts in all 64 cases, and the valet's account of the night fell from 5 of 8 to 4 of 8 by day 7 while the doctor's held at 8 of 8, which is the intended pattern (Fig. 6). With llama3.2:3b the valet's account of the night fell to 1 of 8 by day 7, but the doctor's identity answers also drifted (4 of 8

on day 3): asked about her late husband, the model described a violinist and a gardener rather than the husband killed in 1917 that her memory held. The memory was there; the voice embroidered over it. Selective fallibility in the memory does not guarantee it in the words, and a check on identity facts would be needed for weaker voices.

XI. Discussion

A. What we learned

Decide outside the model, and keep secrets out of the prompt. The clearest result is about architecture rather than memory. When the model holds a secret and is asked to keep it, it gives it away, in about one answer in five (15 of 72). When the engine decides and the secret never reaches the prompt, it cannot leak, whatever the model (0 of 144).

Prompting makes errors a property of the model. Asking a model to "have imperfect memory" produces errors, but which witness errs, and how often, changed completely between two models of the same size. With a memory model deciding, both voices erred in the same band and in the designed order. For a game, this is the difference between a tunable character and an unpredictable one.

The voice is the weak link, and it is model-specific. The words followed the decision in most cases, but both voices turned some told details into first-hand claims, compressed the difference between witnesses, and one embroidered identity facts. These are failures of verbalisation, not of memory, and they are measurable because the decision is logged separately from the words.

Checks are a safety net, not the mechanism. On held-out questions, check-then-retry did not change overall fidelity; it fired on under one answer in twenty. Its value is in guaranteeing hard constraints (no confession, no leaked secret, no anachronism) on the rare answers that break them, not in raising average quality. The decision does the work.

B. Threats to validity

Construct: savings is not recall, and scoring by rules can misjudge a correct answer, as happened with "I have no family of my own"; all answers are released for re-scoring. *Internal*: the authors wrote the probes and rules, although after development and before the final runs. *External*: one case, five characters, two 3B models, English only. *Statistical*: 12 to 15 saves per cell give wide intervals for single probes; we draw conclusions only from differences that hold across both models or are large.

C. What this paper does not show

We have not measured players. Whether people notice these errors, enjoy them, or get better at telling an honest mistake from a lie is the question that matters most for games, and it needs a user study. The interactive house in which the interviews take place is still in development; we used it only as the setting for Study 4's social mechanics and report nothing about its playability.

D. Ethics

Conversational AI can implant false memories in people [23]. Our design does the reverse, implanting them in a fictional character, but the mechanism is the same, and the game makes it visible: an inspector view and the ending show the player how their own questions shaped the testimony. The game depicts coercion, including a confession extracted at gunpoint, as a failure with consequences: a coerced confession leads to a wrongful conviction. All models run locally and no player data leaves the machine.

XII. Conclusion and Future Work

Memory research for language-model agents has mostly asked how to make agents remember better. For a witness in a mystery, the more useful question is how to make them remember the way people do. A small set of standard mechanisms, fitted to human data, gives persistent characters the human error types that accurate memory lacks: false recognition, misinformation, source errors, memory conformity and schema intrusion, while identity stays intact. Deciding those errors outside the language model keeps them tied to the character, keeps secrets out of reach, and makes the remaining failures of the voice measurable. The next steps are a player study comparing fallible and perfect memory, identity and source checks on the spoken answer, and moving interrogative pressure from the source tag to the decision.

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